

Time Guided Lab

Robert Thomas Kohn

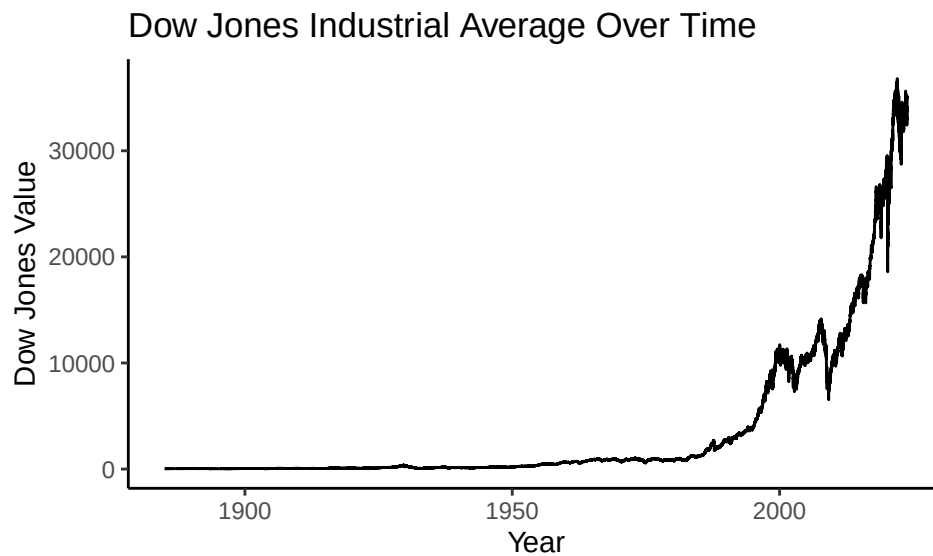
10/19/25

Dow Jones

```
data(DJIA)
head(DJIA)
```

```
## # A tibble: 6 x 2
##   date      value
##   <date>    <dbl>
## 1 1885-02-16  30.9
## 2 1885-02-17  31.3
## 3 1885-02-18  31.5
## 4 1885-02-19  31.7
## 5 1885-02-20  31.4
## 6 1885-02-21  31.9
```

```
gf_line(value ~ date, data = DJIA) %>% gf_labs(title = "Dow Jones Industrial Average Over Time", x = "Year")
```



```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2000)) %>% gf_labs(title = "Dow Jones AVG Since 2000")
```

Dow Jones AVG Since 2000



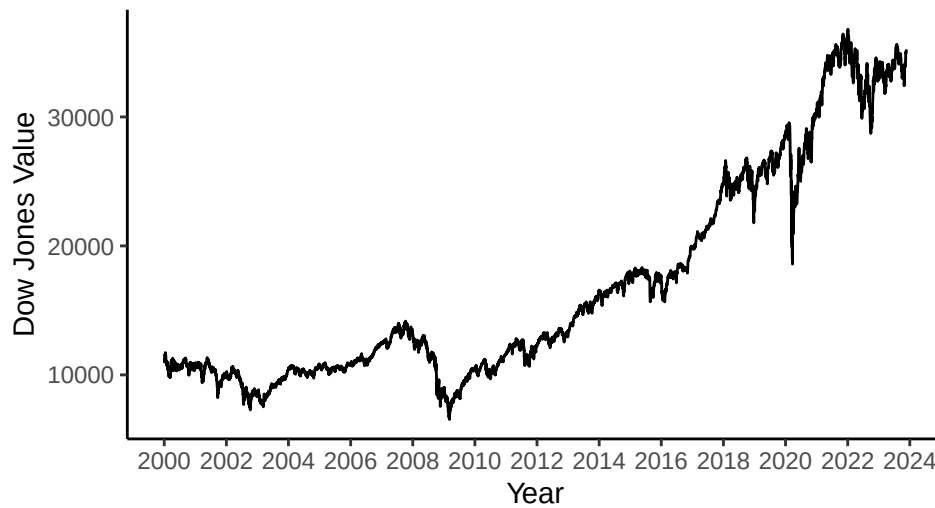
```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2000)) %>%gf_refine(scale_x_date(date_breaks =
```

Dow Jones with 2-Year Breaks



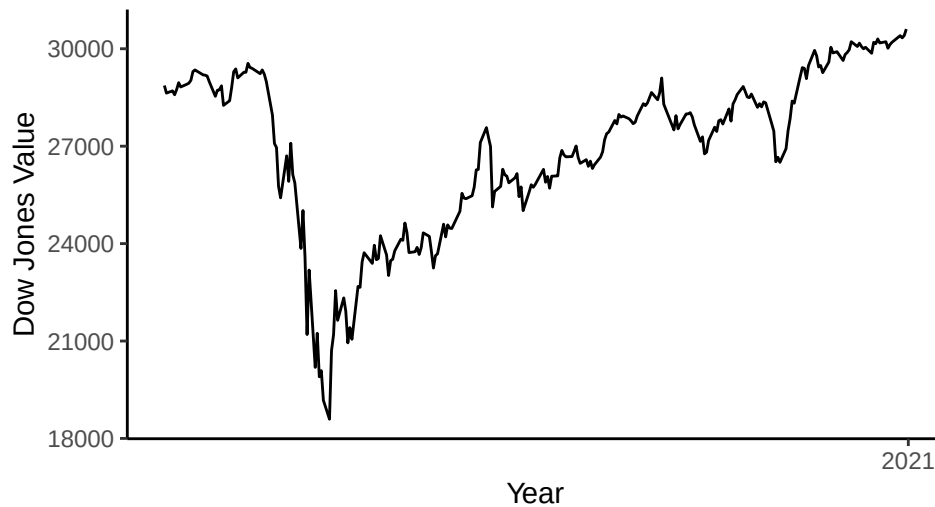
```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2000)) %>%gf_refine(scale_x_date(date_breaks =
```

Dow Jones with Better Labels

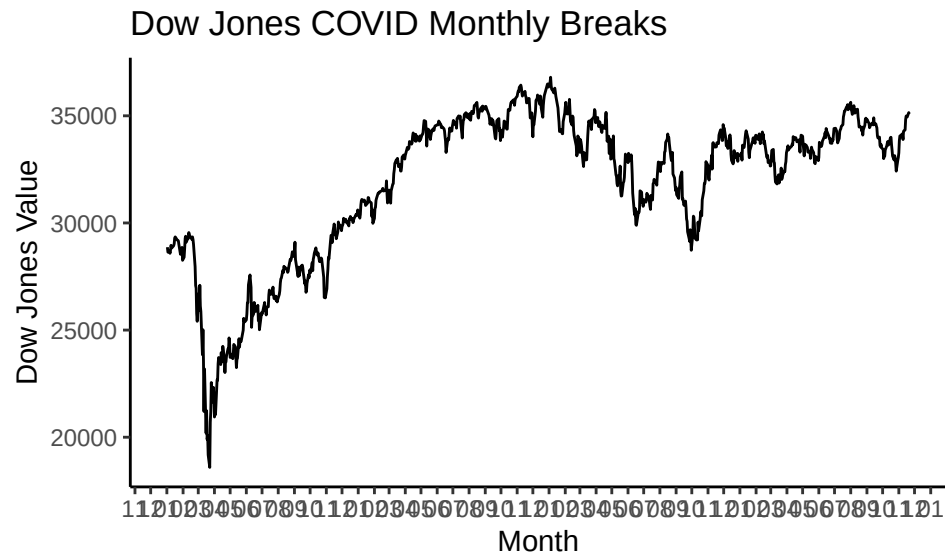


```
gf_line(value ~ date, data = subset(DJIA, year(date) == 2020)) %>%gf_refine(scale_x_date(date_breaks =
```

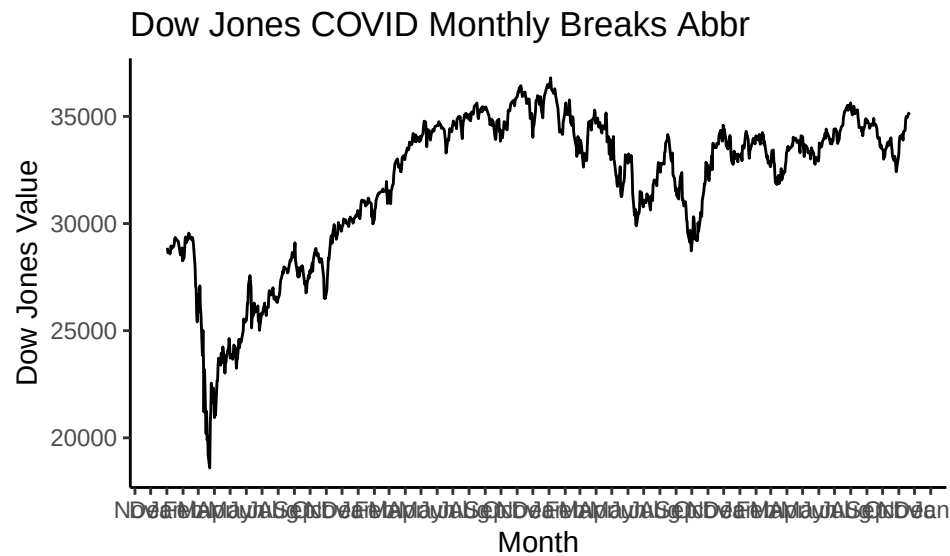
Dow Jones COVID



```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2020)) %>%
  gf_refine(scale_x_date(date_breaks = "1 month", date_labels = "%m")) %>%
  gf_labs(title = "Dow Jones COVID Monthly Breaks", x = "Month", y = "Dow Jones Value")
```



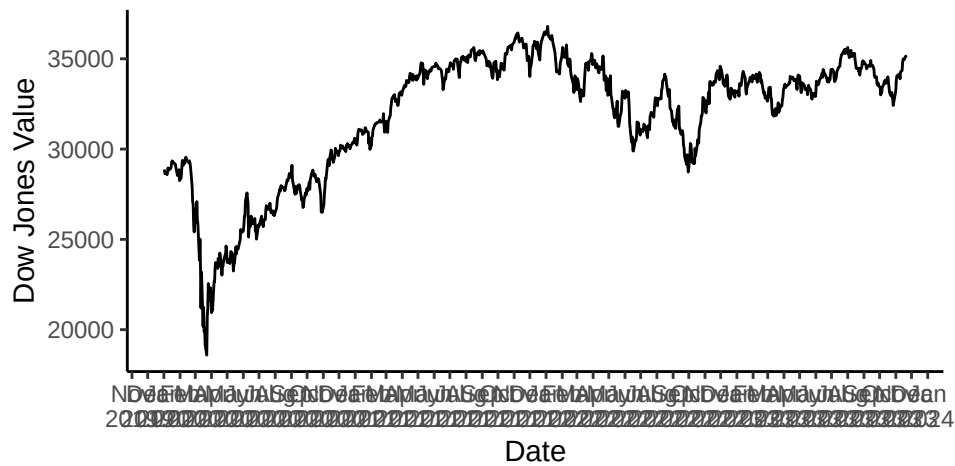
```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2020)) %>%
  gf_refine(scale_x_date(date_breaks = "1 month", date_labels = "%b")) %>%
  gf_labs(title = "Dow Jones COVID Monthly Breaks Abbr", x = "Month", y = "Dow Jones Value")
```



```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2020)) %>% gf_refine(scale_x_date(date_breaks =
```

Dow Jones Trends Since 2000

Monthly view with year labels

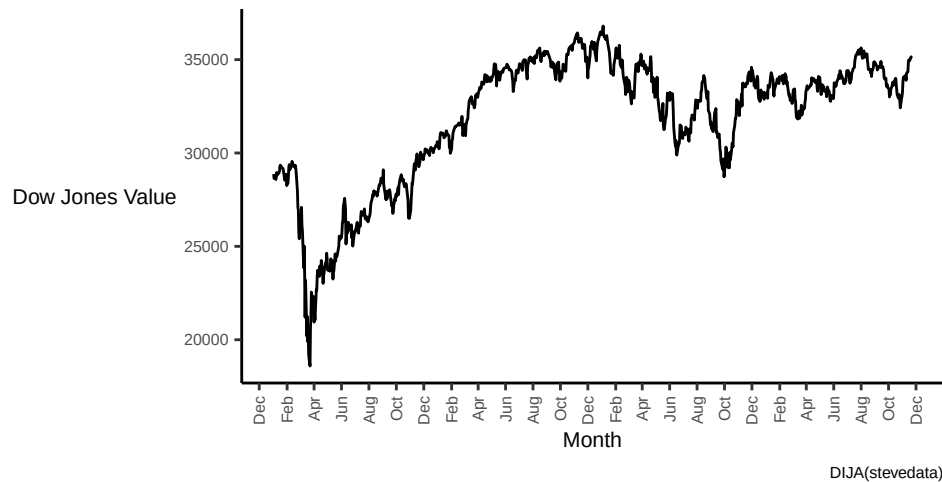


#Final

```
gf_line(value ~ date, data = subset(DJIA, year(date) >= 2020)) %>% gf_refine(scale_x_date(date_breaks =
```

Dow Jones increased rapidly as COVID progressed

Bi-Monthly Dow Jones Value(2020)



DJIA(stevedata)

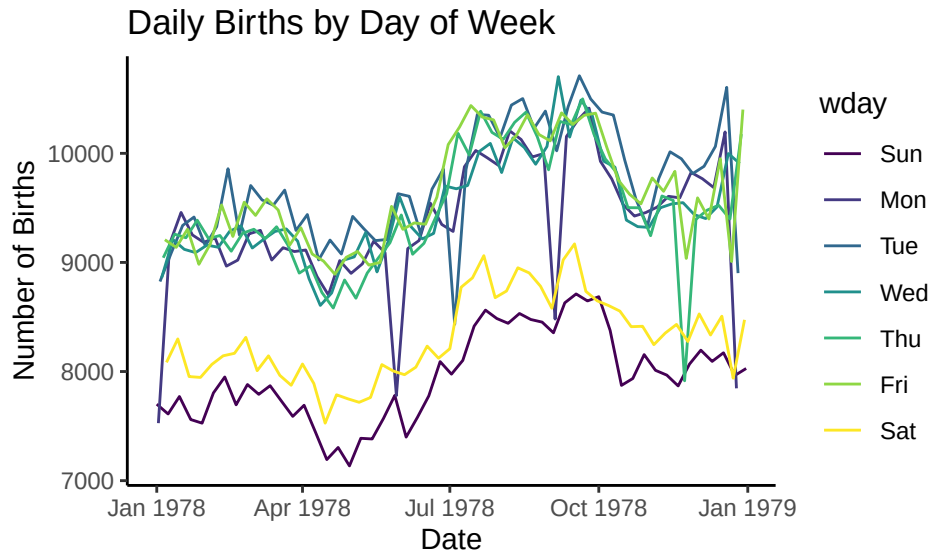
BIRTHS SECTION

```
data(Births78)
```

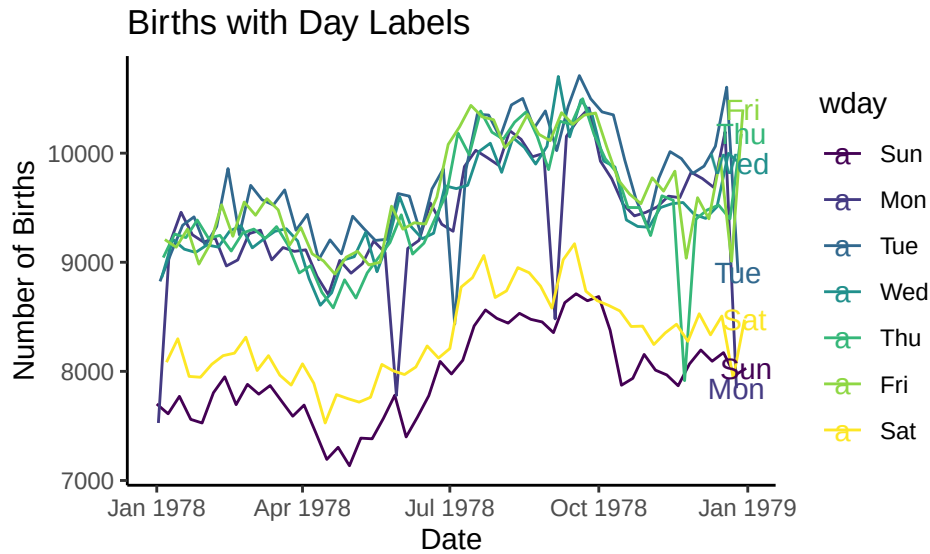
```
head(Births78)
```

```
##      date births wday year month day_of_year day_of_month day_of_week
## 1 1978-01-01   7701  Sun 1978     1           1           1           1
## 2 1978-01-02   7527  Mon 1978     1           2           2           2
## 3 1978-01-03   8825  Tue 1978     1           3           3           3
## 4 1978-01-04   8859  Wed 1978     1           4           4           4
## 5 1978-01-05   9043  Thu 1978     1           5           5           5
## 6 1978-01-06   9208  Fri 1978     1           6           6           6
```

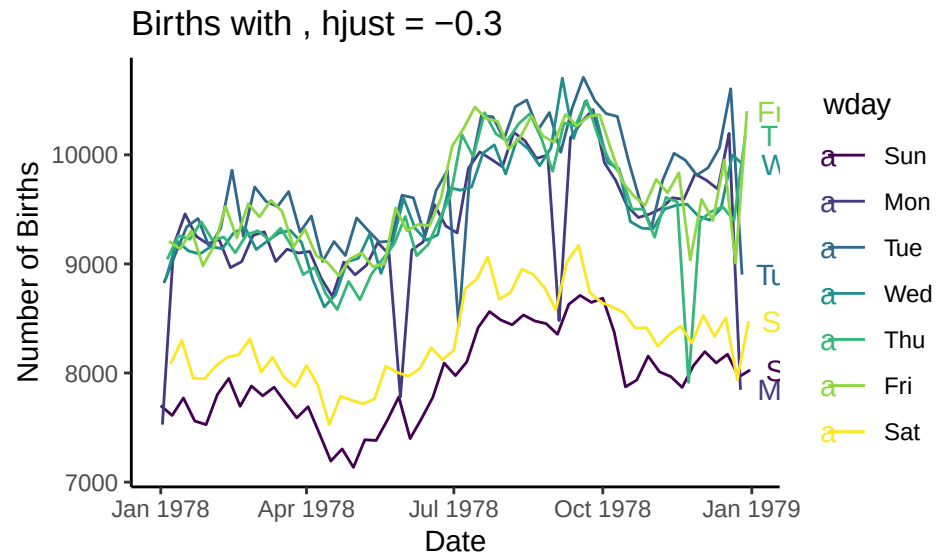
```
gf_line(births ~ date, data = Births78, color = ~wday) %>%gf_labs(title = "Daily Births by Day of Week"
```



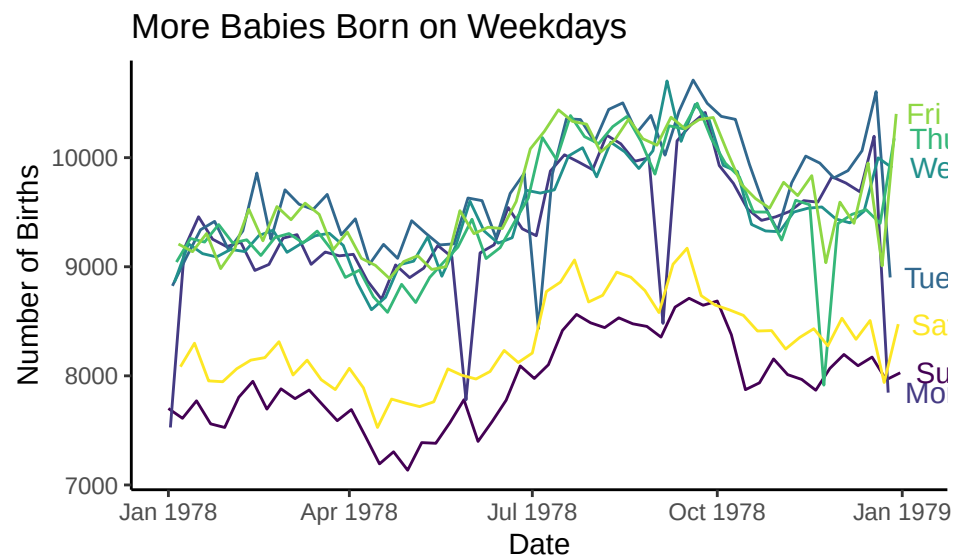
```
gf_line(births ~ date, data = Births78, color = ~wday) %>%gf_text(births ~ date, data = subset(Births78
```



```
gf_line(births ~ date, data = Births78, color = ~wday) %>%gf_text(births ~ date, data = subset(Births78
```



```
gf_line(births ~ date, data = Births78, color = ~wday) %>% gf_text(births ~ date, data = subset(Births78
```

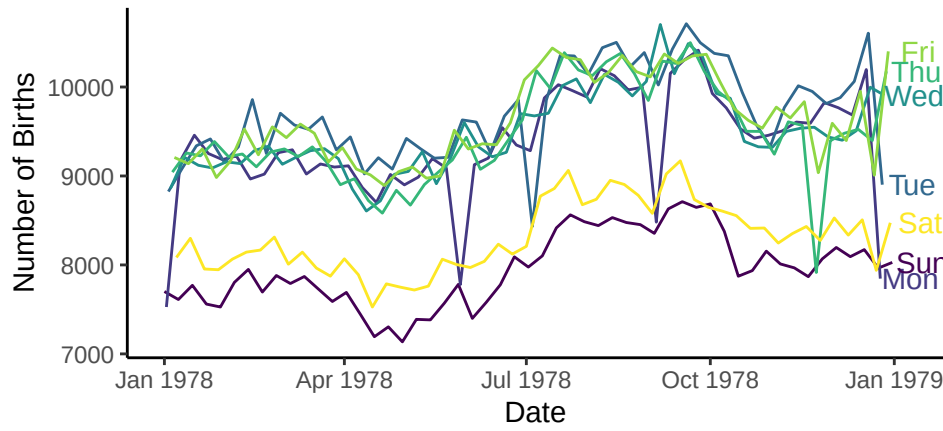


#Final

```
gf_line(births ~ date, data = Births78, color = ~wday) %>% gf_text(births ~ date, data = subset(Births78
```

More Babies Born on Weekdays

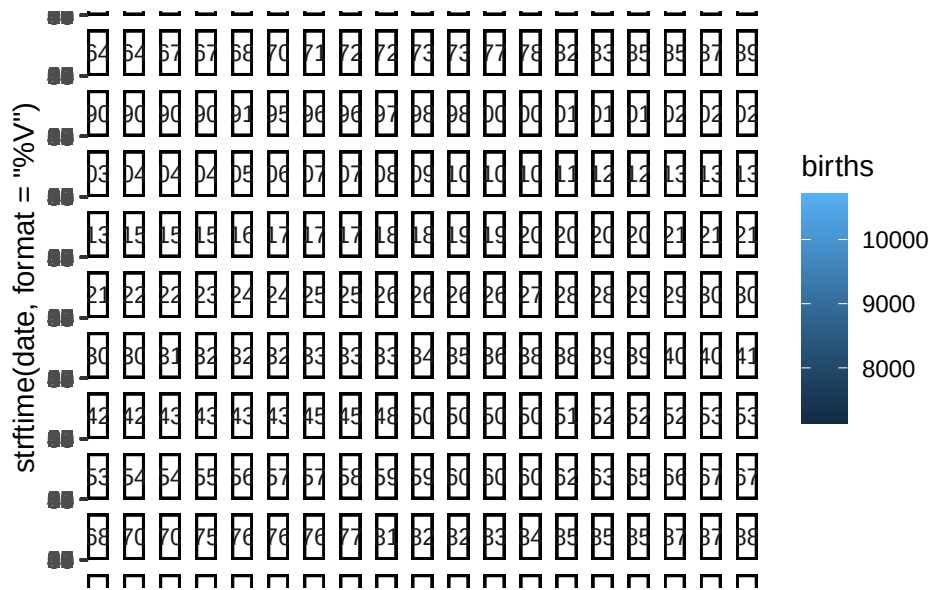
Birthrate by Weekday



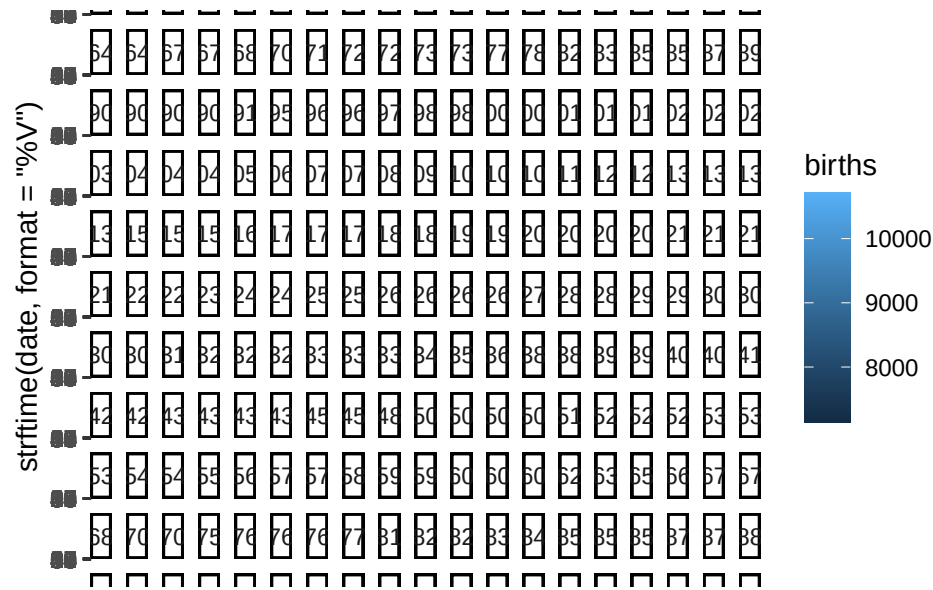
Births78(Lock5Data)

CALENDAR VIEW SECTION

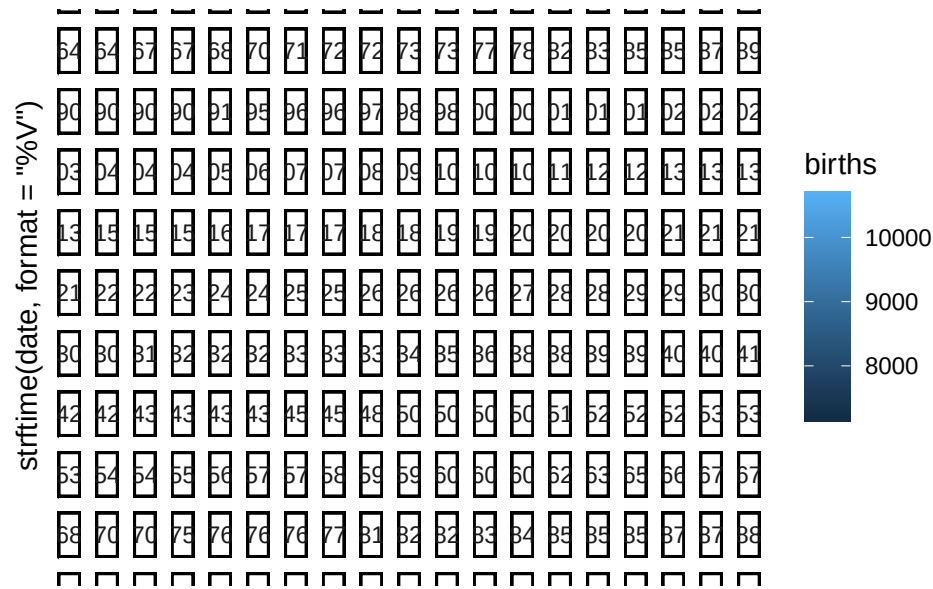
```
gf_tile( strftime(date, format = "%V") ~ weekdays(date) , data = Births78 , fill = ~births ) %>% gf_fac
```



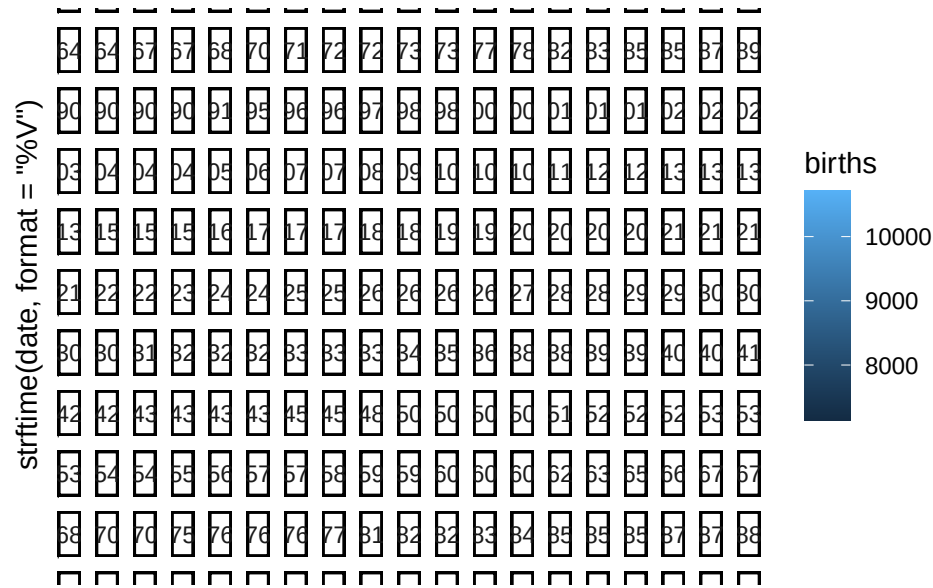
```
gf_tile( strftime(date, format = "%V") ~ weekdays(date) , data = Births78 , fill = ~births ) %>% gf_fac
```



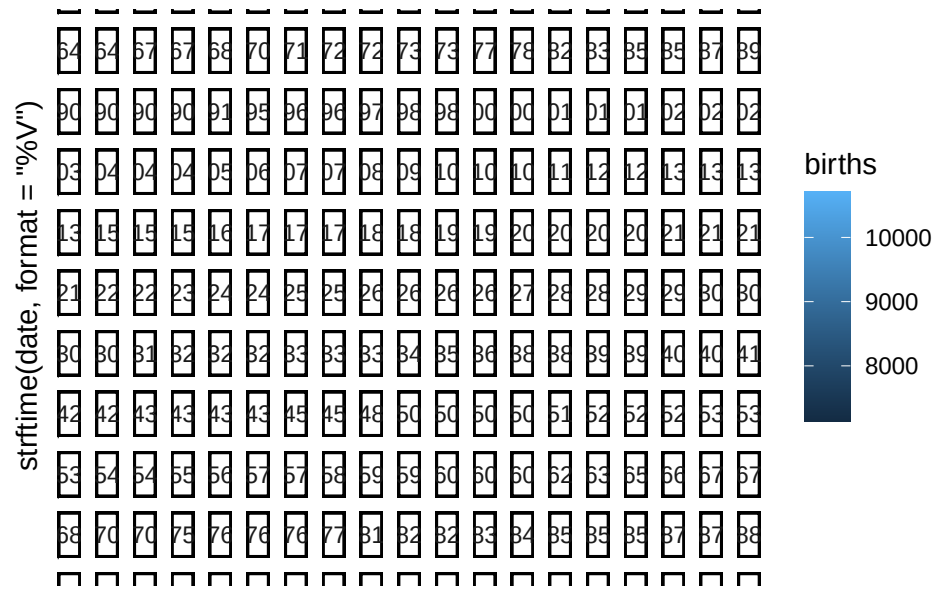
```
gf_tile( strftime(date, format = "%V") ~ weekdays(date) , data = Births78 , fill = ~births ) %>% gf_facet_wrap( ~
```



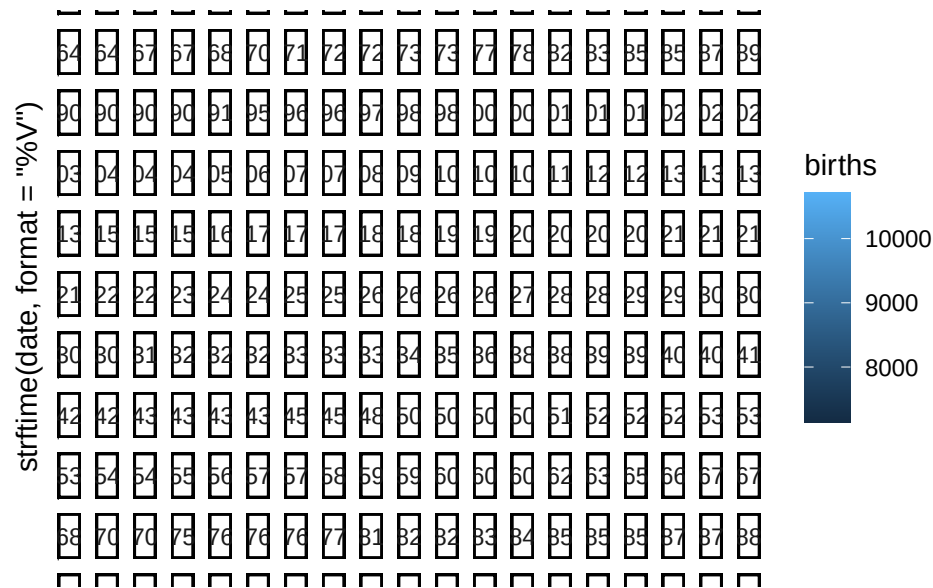
```
gf_tile( strftime(date, format = "%V") ~ wday , data = Births78 , fill = ~births ) %>% gf_facet_wrap( ~
```



```
gf_tile( strftime(date, format = "%V") ~ wday , data = Births78 , fill = ~births ) %>% gf_facet_wrap( ~
```

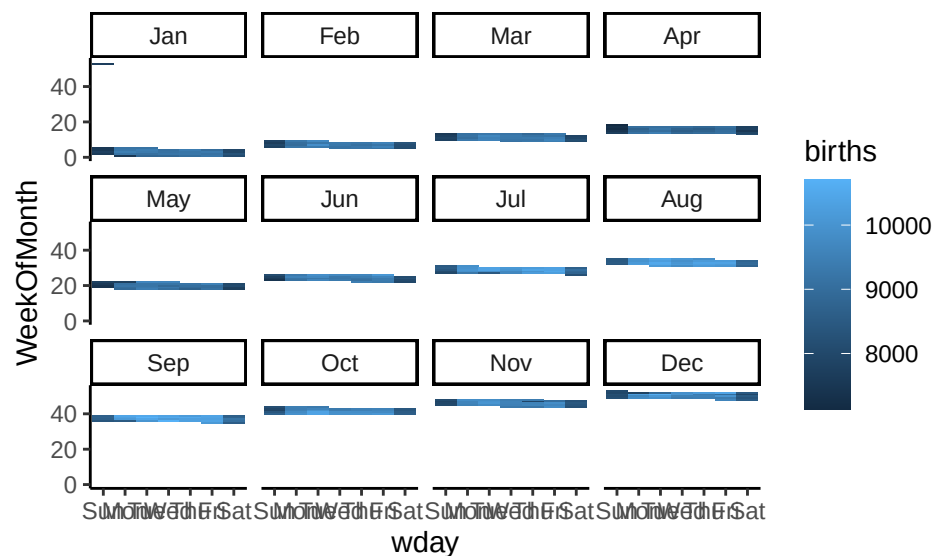


```
gf_tile( strftime(date, format = "%V") ~ wday , data = Births78 , fill = ~births ) %>% gf_facet_wrap( ~
```

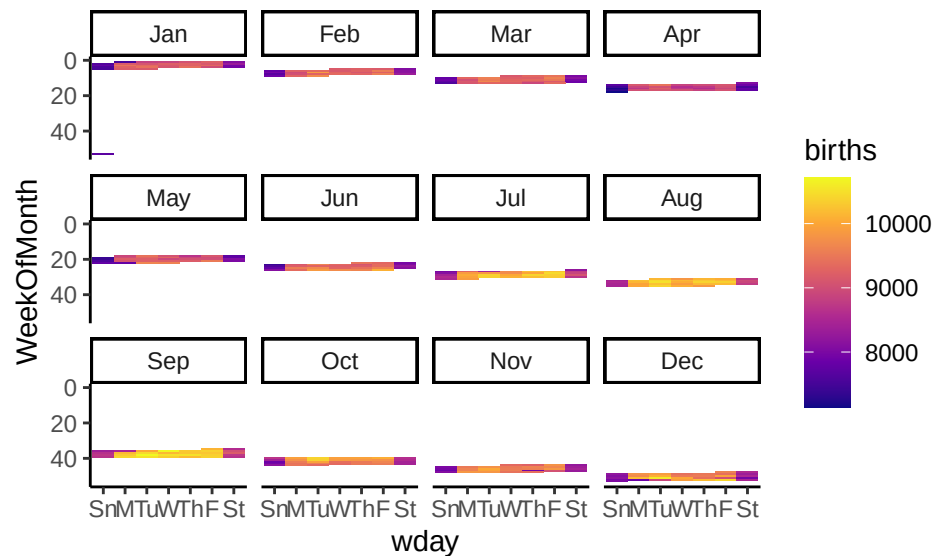


```
Births_calendar <- Births78 %>% mutate(WeekOfMonth = ifelse(weekdays(date) == "Sunday", as.numeric(strftime(date, format = \"%V\") - 1), 0))
```

```
gf_tile(WeekOfMonth ~ wday, data = Births_calendar, fill = ~births) %>% gf_facet_wrap(~month(date, abbr = TRUE))
```



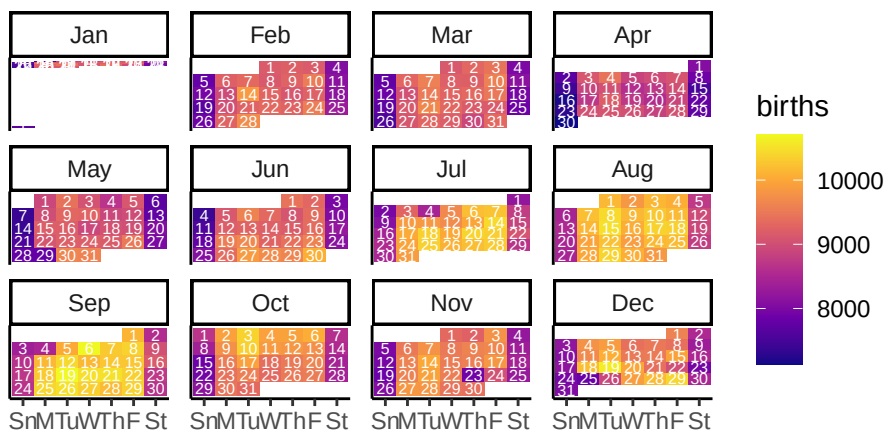
```
gf_tile(WeekOfMonth ~ wday, data = Births_calendar, fill = ~births) %>% gf_facet_wrap(~month(date, abbr = TRUE))
```



```
gf_tile(WeekOfMonth ~ wday, data = Births_calendar, fill = ~births) %>% gf_facet_wrap(~month(date, abbr = TRUE))
```

Fewer Babies Born on Holidays & Weekends

Births Calendar View 1978



HUMIDITY BOXPLOTS SECTION

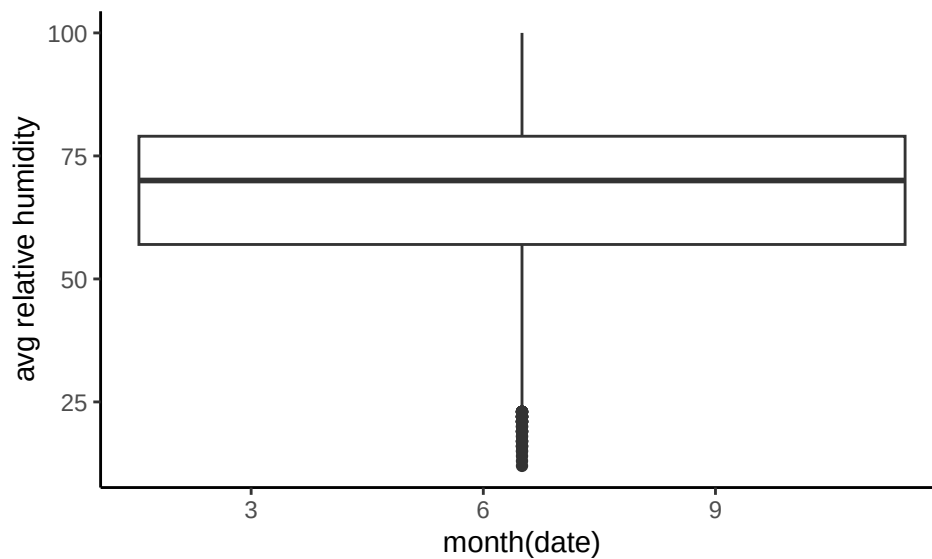
```
data(Weather)
head(Weather)
```

```
## # A tibble: 6 x 25
##   city    date      year month   day high_temp avg_temp low_temp high_dewpt
##   <chr>   <date>    <dbl> <int> <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 Auckland 2016-01-01 2016     1     1      68      65      62      64
## 2 Auckland 2016-01-02 2016     1     2      68      66      64      64
## 3 Auckland 2016-01-03 2016     1     3      77      72      66      70
## 4 Auckland 2016-01-04 2016     1     4      73      66      60      66
## 5 Auckland 2016-01-05 2016     1     5      69      62      55      55
## 6 Auckland 2016-01-06 2016     1     6      69      63      57      54
```

```
## # i 16 more variables: avg_dewpt <dbl>, low_dewpt <dbl>, high_humidity <dbl>,  
## #   avg_humidity <dbl>, low_humidity <dbl>, high_hg <dbl>, avg_hg <dbl>,  
## #   low_hg <dbl>, high_vis <dbl>, avg_vis <dbl>, low_vis <dbl>,  
## #   high_wind <dbl>, avg_wind <dbl>, low_wind <dbl>, precip <chr>, events <chr>
```

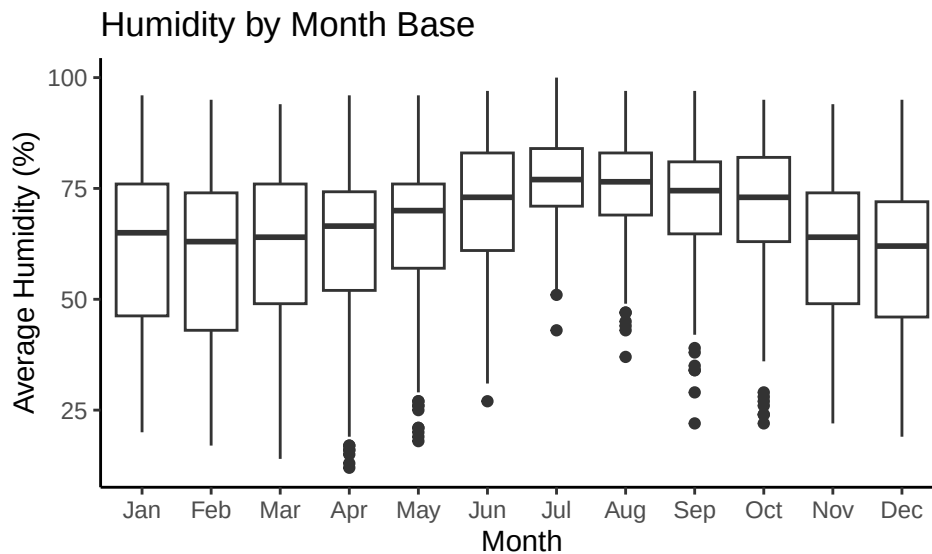
```
gf_boxplot(avg_humidity ~ month(date), data = Weather)
```

```
## Warning: Continuous x aesthetic  
## i did you forget `aes(group = ...)`?
```



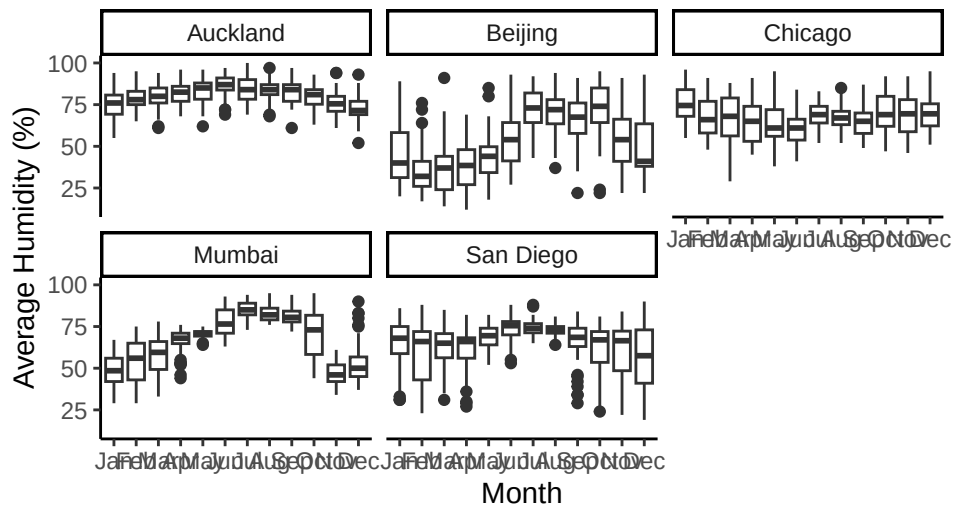
```
WeatherWMonth <- Weather %>% mutate(Month = month(date, abbr = TRUE, label = TRUE))
```

```
gf_boxplot(avg_humidity ~ Month, data = WeatherWMonth) %>% gf_labs(title = "Humidity by Month Base", x = Month)
```



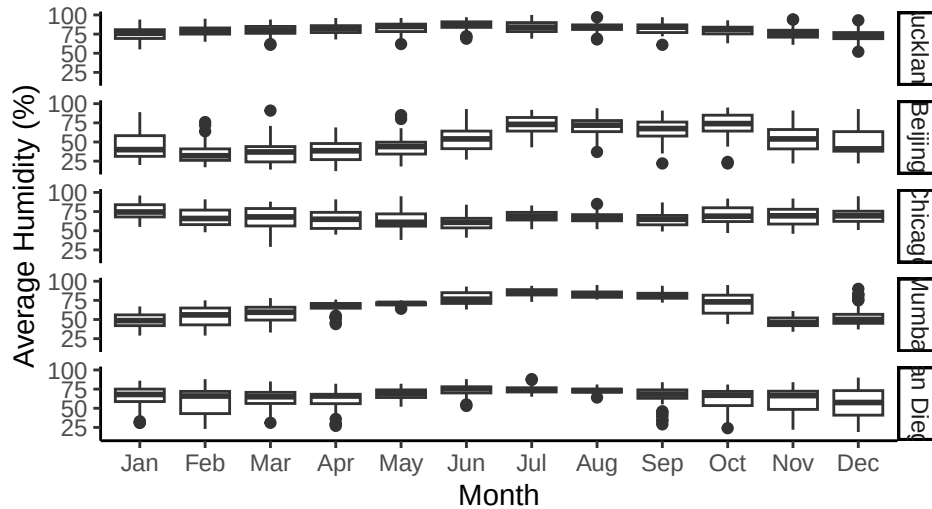
```
gf_boxplot(avg_humidity ~ Month, data = WeatherWMonth) %>% gf_facet_wrap(~city) %>% gf_theme(strip.text = FALSE)
```

Humidity Patterns by City

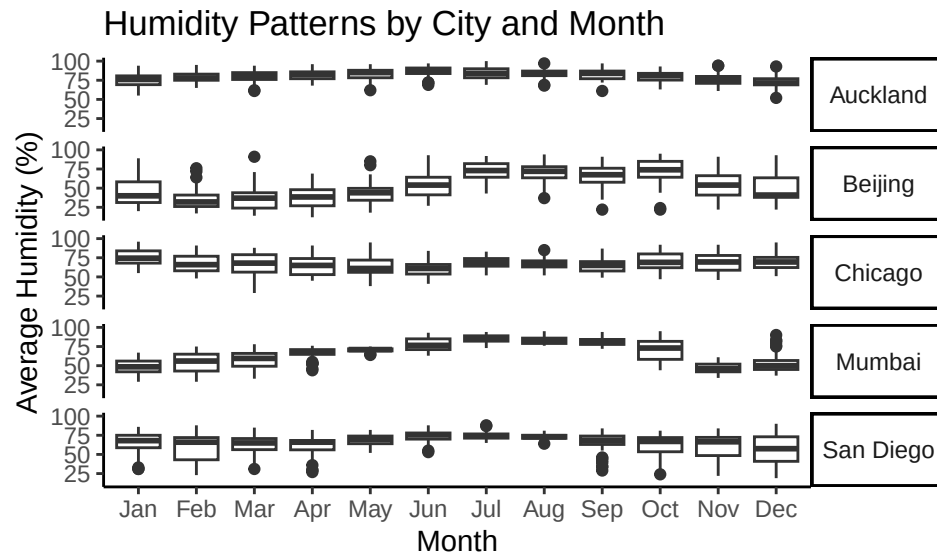


```
gf_boxplot(avg_humidity ~ Month, data = WeatherWMonth) %>% gf_facet_wrap(~city, ncol = 1, strip.position = "right")
```

Humidity Patterns by City and Month

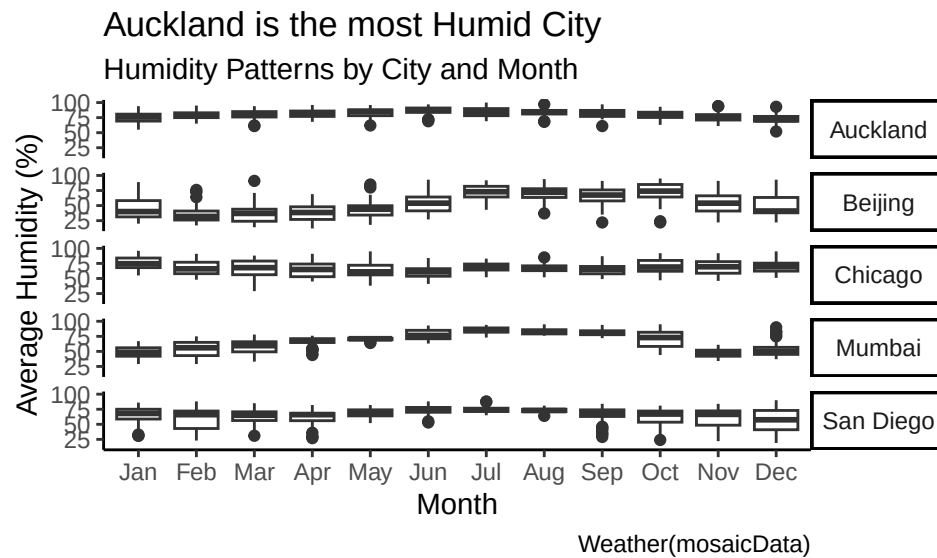


```
gf_boxplot(avg_humidity ~ Month, data = WeatherWMonth) %>% gf_facet_wrap(~city, ncol = 1, strip.position = "right")
```



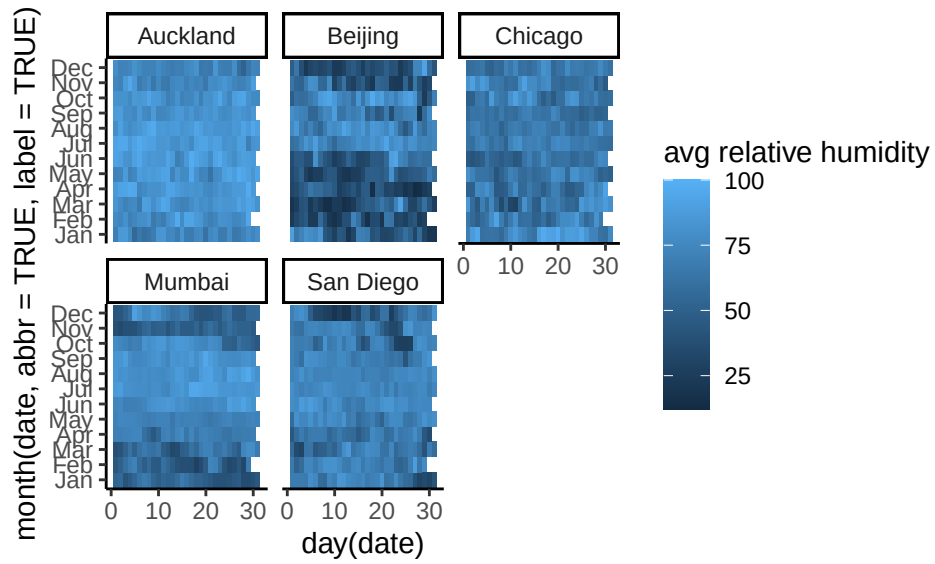
#Final

```
gf_boxplot(avg_humidity ~ Month, data = WeatherWMonth) %>% gf_facet_wrap(~city, ncol = 1, strip.position = "right")
```

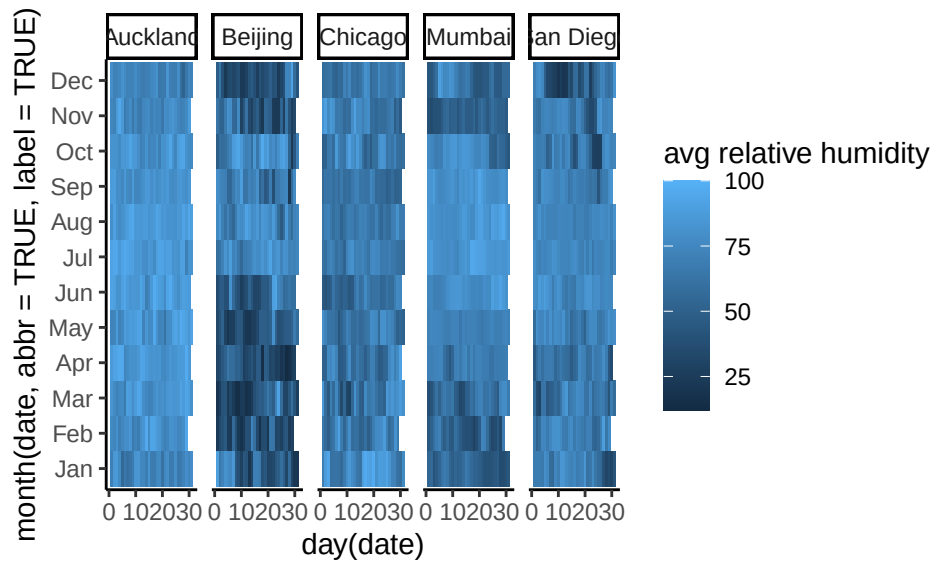


HUMIDITY TILE GRAPH SECTION

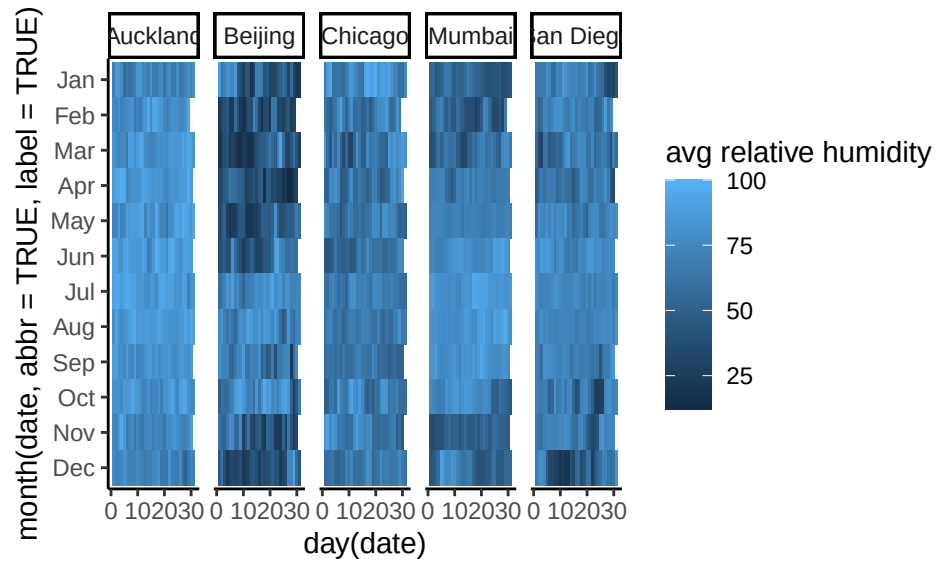
```
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>% gf_facet_wrap(~city, ncol = 1, strip.position = "right")
```



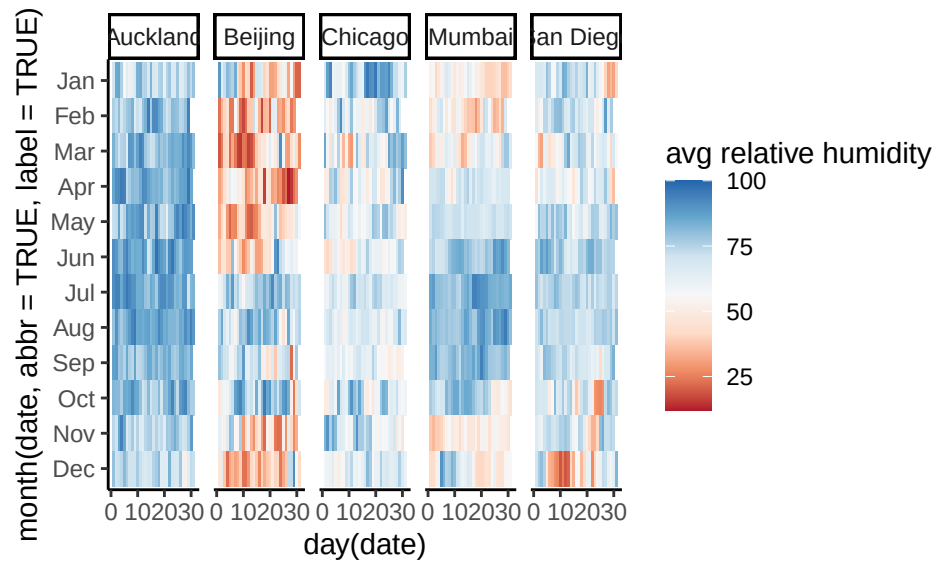
```
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>% gf_tile
```



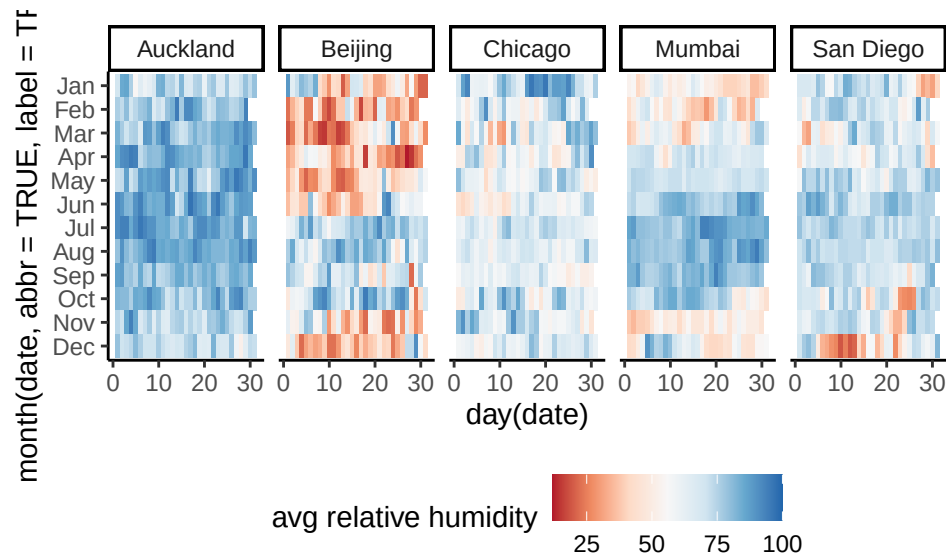
```
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>%
```



```
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>% g
```



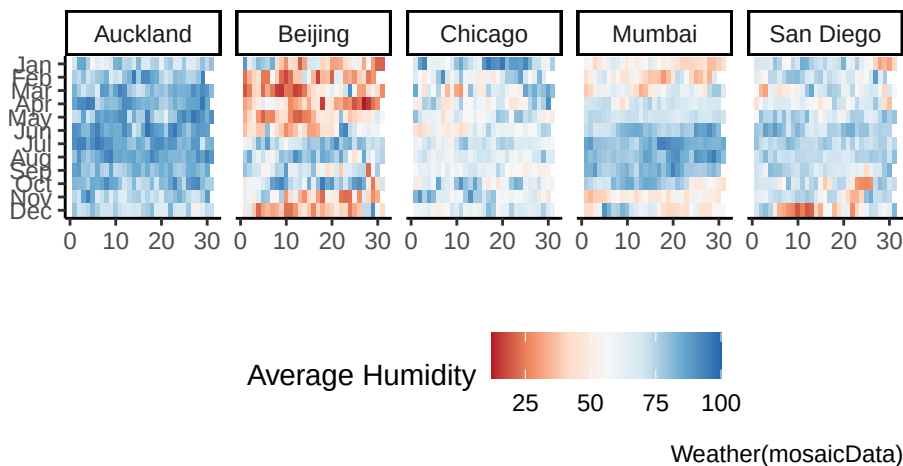
```
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>% g
```



```
#Final
gf_tile(month(date, abbr = TRUE, label = TRUE) ~ day(date), data = Weather, fill = ~avg_humidity) %>% g
```

Beijing has a Lower Humidity than other Cities

Average Humidity Patterns by Day and Month



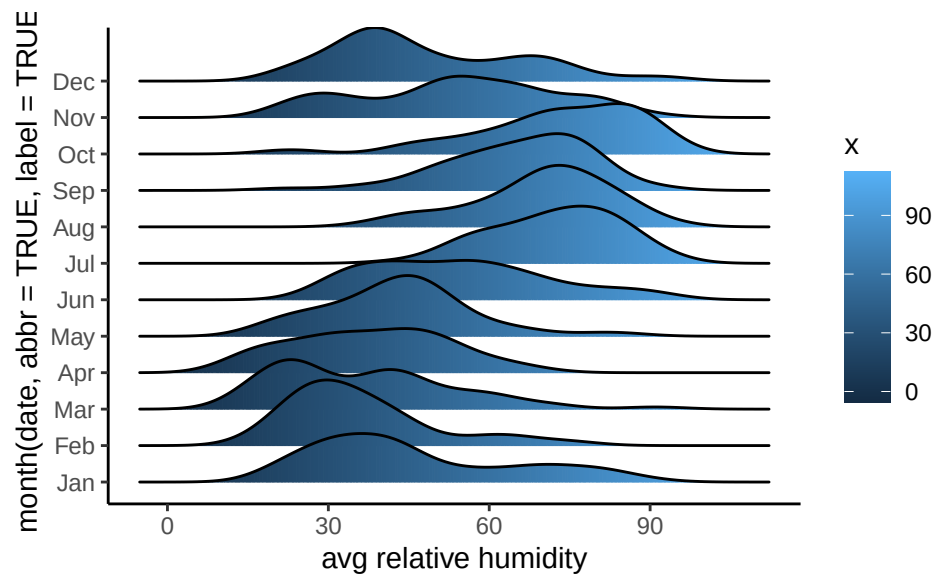
GRADIENT RIDGES SECTION

```
BeijingWeather <- subset(Weather, city == "Beijing")

gf_density_ridges_gradient(month(date, abbr = TRUE, label = TRUE) ~ avg_humidity, data = BeijingWeather)

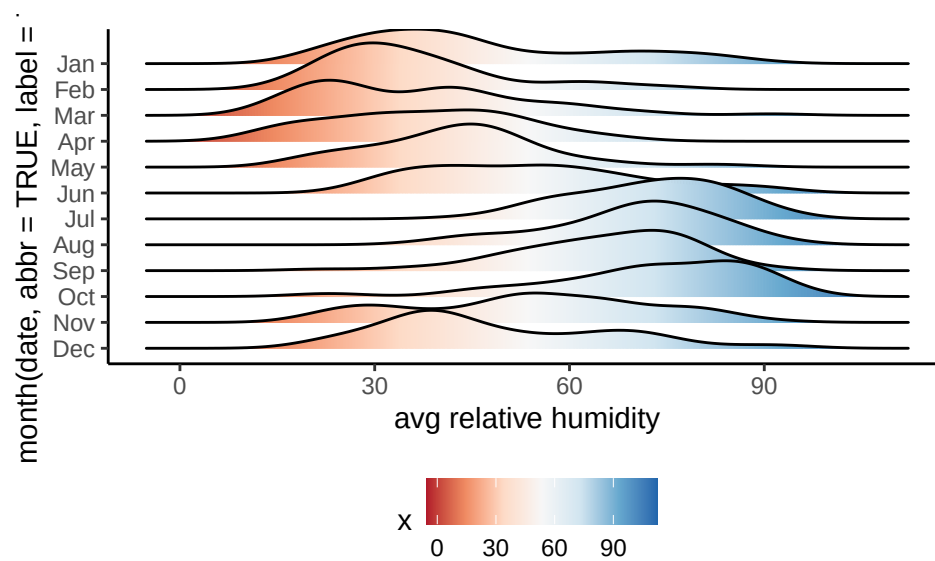
## Warning: `stat(x)` was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(x)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Picking joint bandwidth of 5.73
```



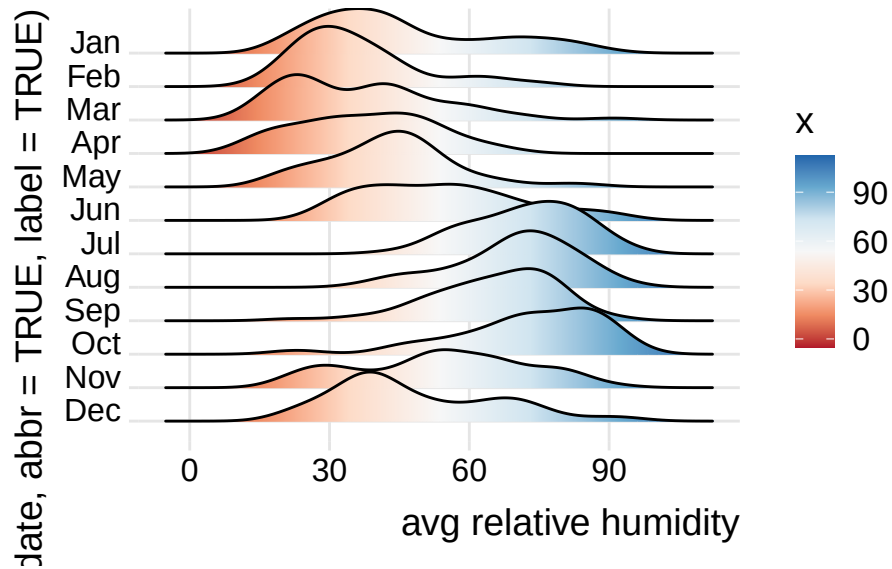
```
gf_density_ridges_gradient(month(date, abbr = TRUE, label = TRUE) ~ avg_humidity, data = BeijingWeather)
```

```
## Picking joint bandwidth of 5.73
```



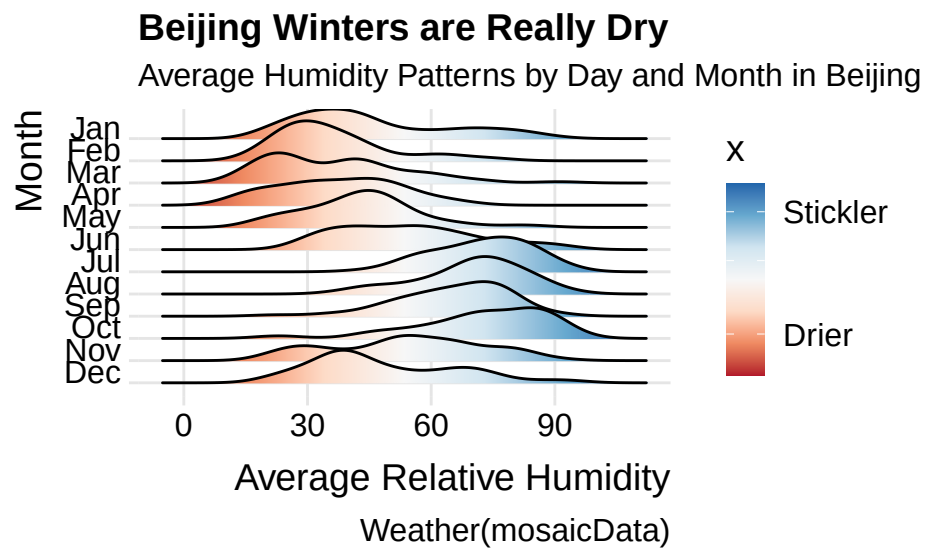
```
gf_density_ridges_gradient(month(date, abbr = TRUE, label = TRUE) ~ avg_humidity, data = BeijingWeather)
```

```
## Picking joint bandwidth of 5.73
```



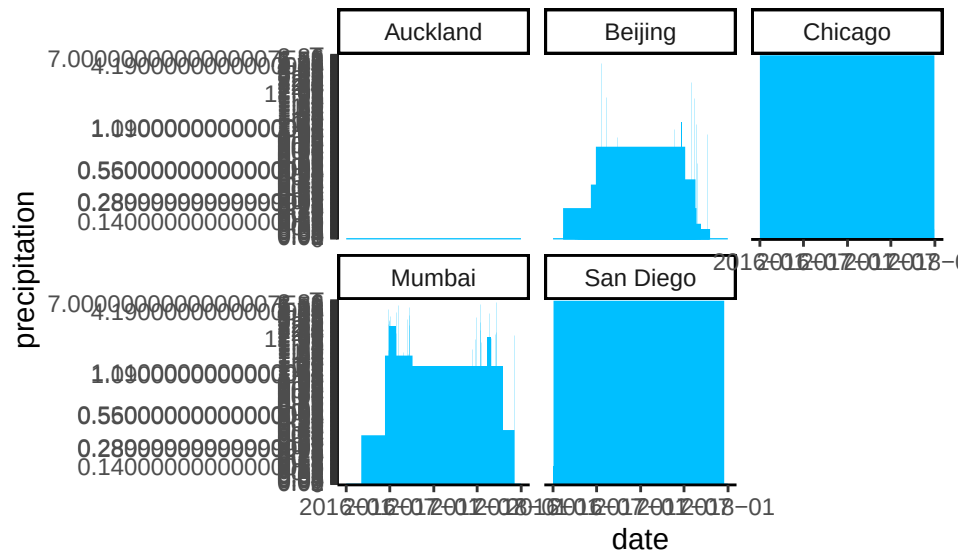
```
#final
gf_density_ridges_gradient(month(date, abbr = TRUE, label = TRUE) ~ avg_humidity, data = BeijingWeather)
```

```
## Picking joint bandwidth of 5.73
```



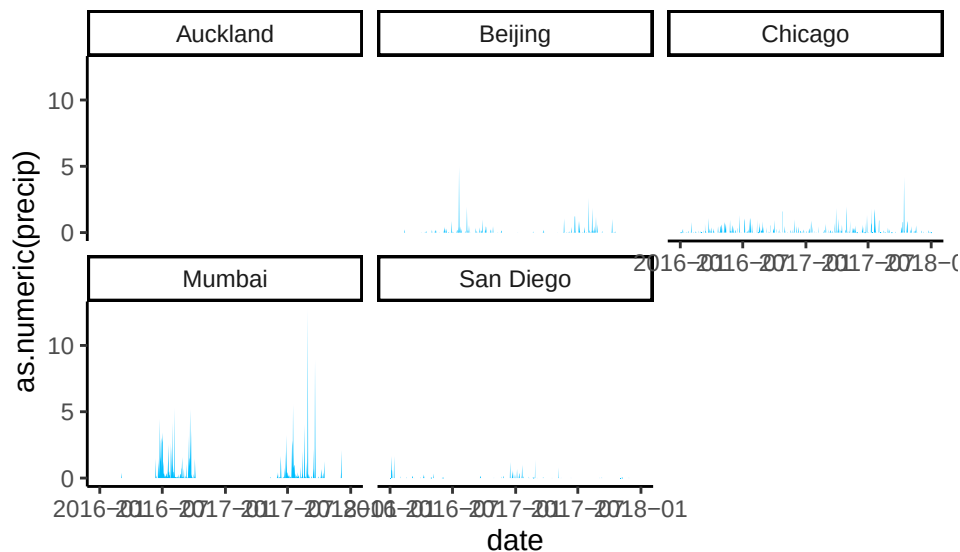
PRECIPITATION AREA PLOT SECTION

```
gf_area(precip ~ date, data = Weather, fill = "deepskyblue") %>% gf_facet_wrap(~city)
```



```
gf_area(as.numeric(precip) ~ date, data = Weather, fill = "deepskyblue") %>%gf_facet_wrap(~city)
```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```



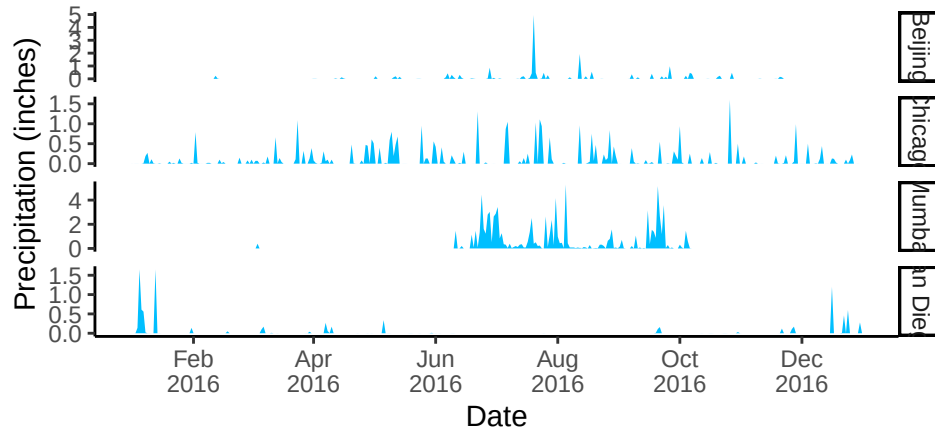
```
WeatherSubset <- subset(Weather, city != "Auckland" & year(date) == 2016)
WSC <- WeatherSubset %>%mutate(precip_numeric = ifelse(precip == "T", 0.001, as.numeric(precip)))
```

```
## Warning: There was 1 warning in `mutate()`.
## i In argument: `precip_numeric = ifelse(precip == "T", 0.001,
##   as.numeric(precip))`.
## Caused by warning in `ifelse()`:
## ! NAs introduced by coercion
```

```
#Final
gf_area(precip_numeric ~ date, data = WSC, fill = "deepskyblue") %>%gf_facet_wrap(~city, ncol = 1, strip
```

Chicago has precipitation all year round

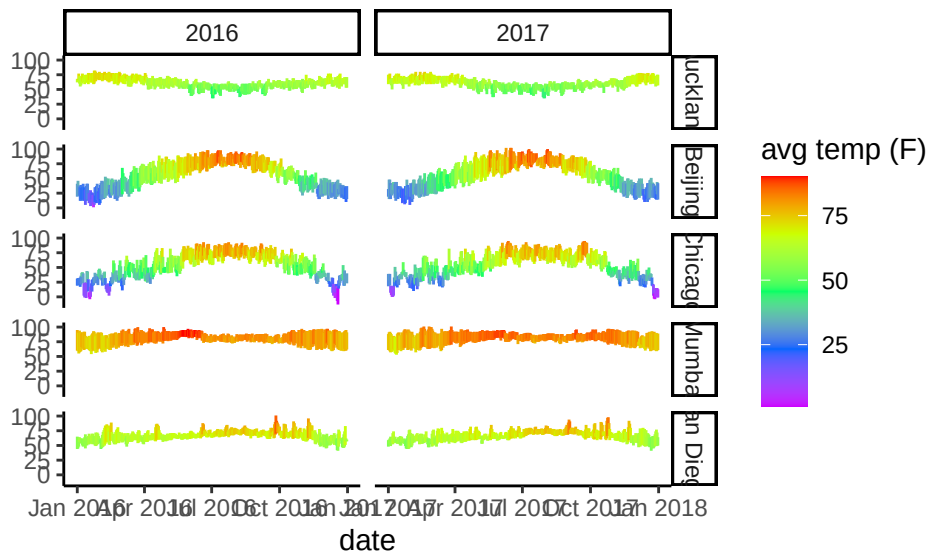
Precipitation Patterns by City



Weather(mosaicData)

TEMPERATURE LINERANGE PLOT SECTION

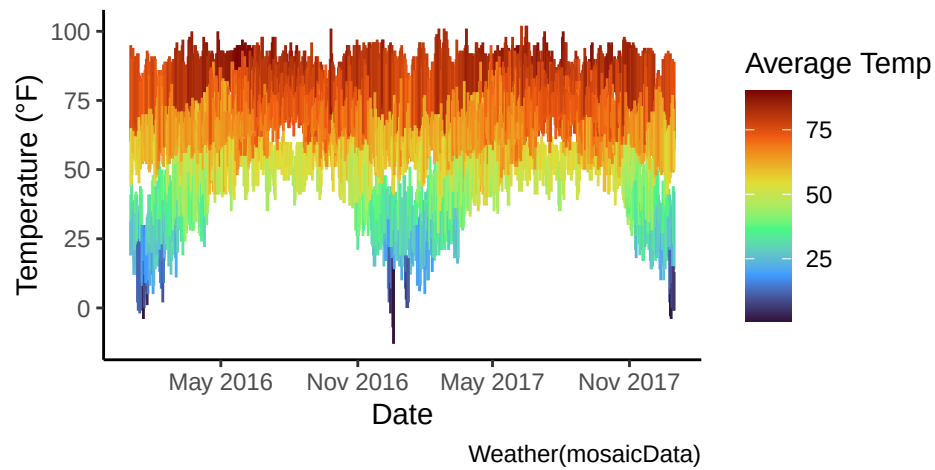
```
gf_linerange(low_temp + high_temp ~ date, data = Weather, color = ~avg_temp) %>%gf_refine(scale_color_g
```



```
gf_linerange(low_temp + high_temp ~ date, data = Weather, color = ~avg_temp) %>%gf_refine(scale_color_v
```

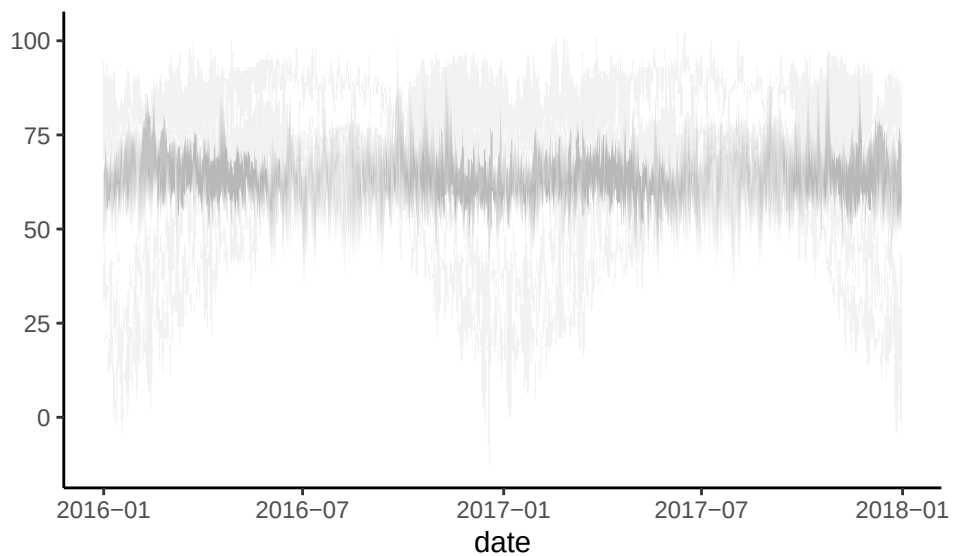
Chicago has Rapid Temperature Swings

Daily Temperature Ranges

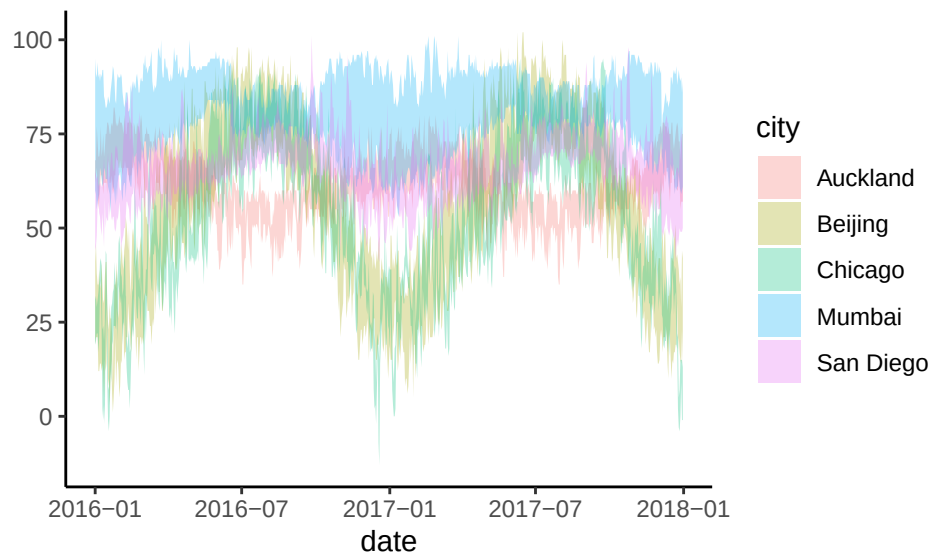


TEMPERATURE RIBBON PLOT SECTION

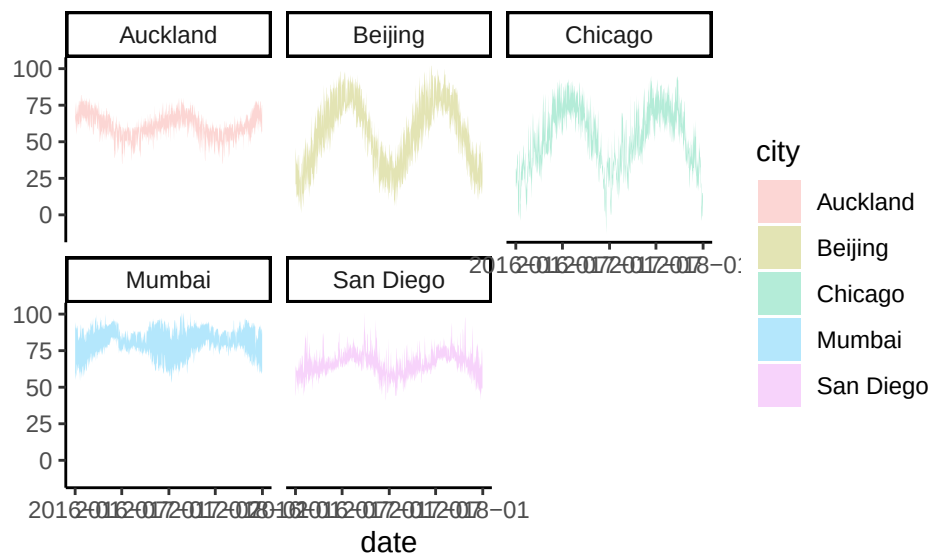
```
gf_ribbon( low_temp + high_temp ~ date , data = Weather)
```



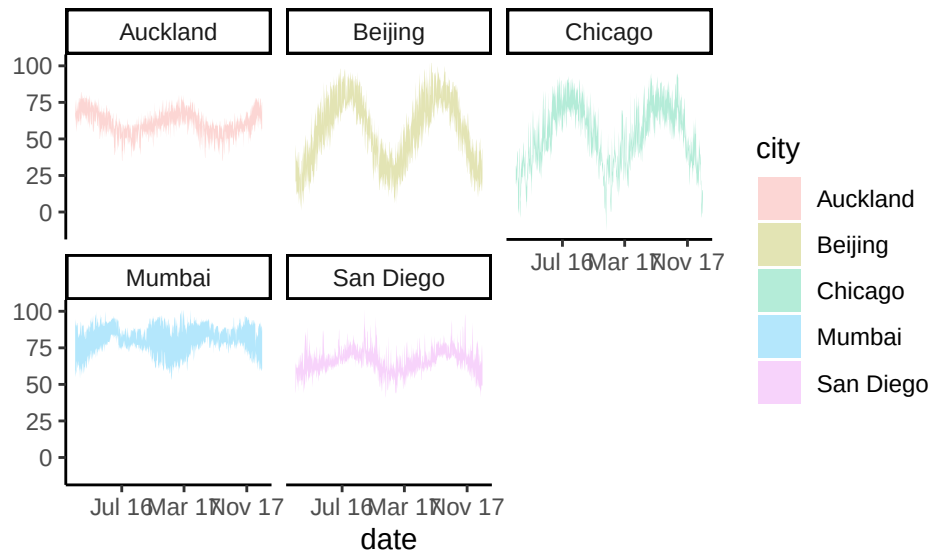
```
gf_ribbon( low_temp + high_temp ~ date , data = Weather, fill=~city)
```



```
gf_ribbon( low_temp + high_temp ~ date , data = Weather, fill=~city)%>%gf_facet_wrap(~city)
```



```
gf_ribbon( low_temp + high_temp ~ date , data = Weather, fill=~city)%>%gf_facet_wrap(~city)%>%gf_refine
```

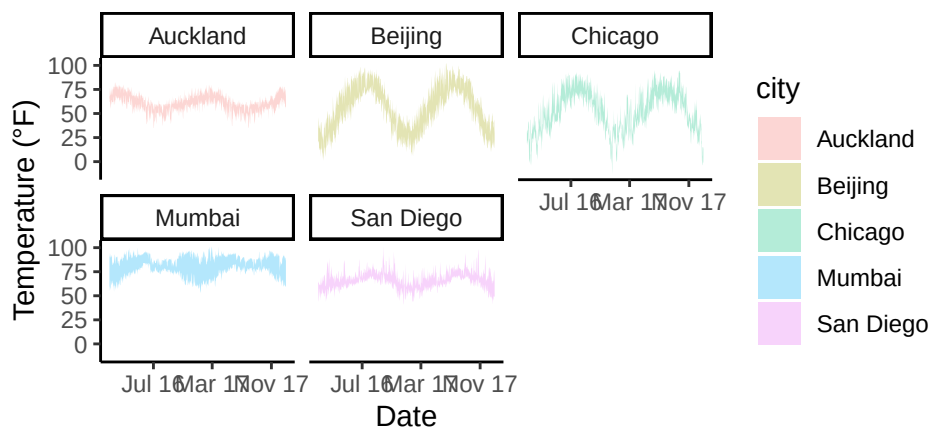


#Final

```
gf_ribbon( low_temp + high_temp ~ date , data = Weather, fill=~city)%>%gf_facet_wrap(~city)%>%gf_refine
```

Chicago and Beijing have Rapid Temperature Swings

Daily Temperature Ranges by City



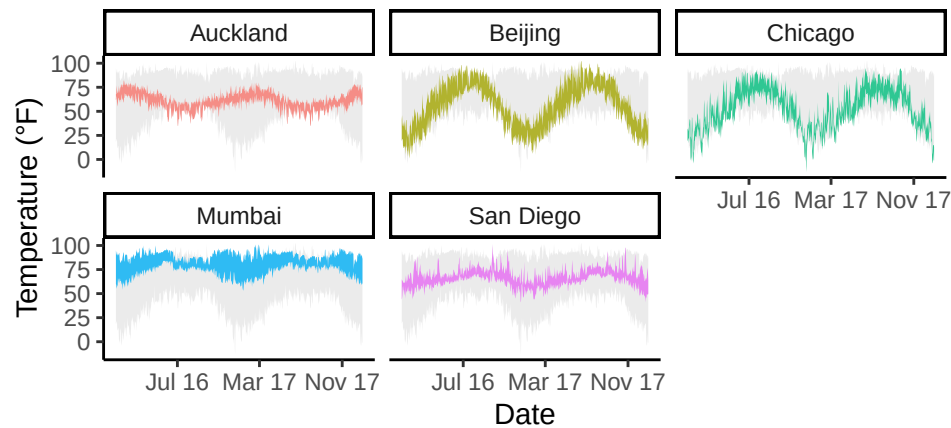
Weather(mosaicData)

```
HighLow <- Weather %>% group_by(date) %>%mutate(lowest = min(low_temp),highest = max(high_temp))
```

```
gf_ribbon(lowest + highest ~ date, data = HighLow, fill = "gray85", alpha = 0.5) %>% gf_ribbon(low_temp
```

San Diego has the Most Average Temperature

Temperature Ranges by City

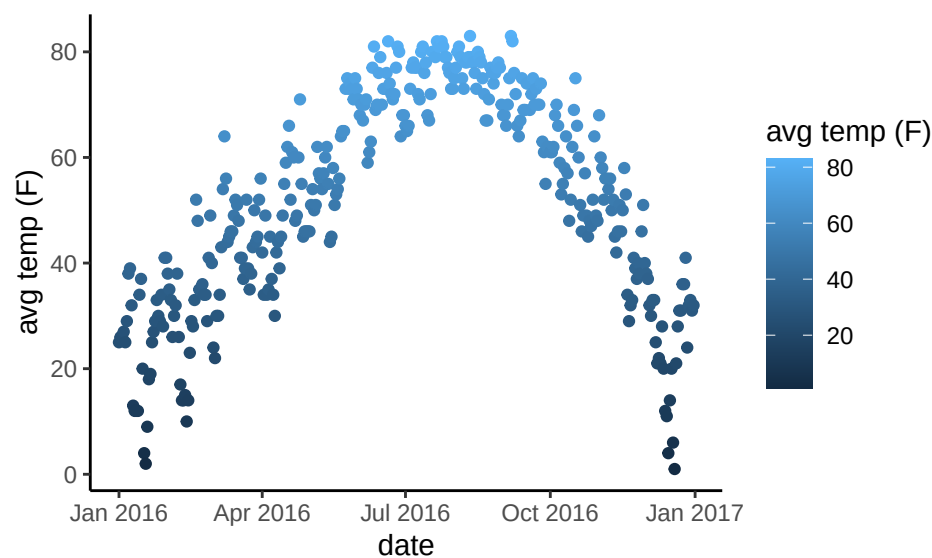


Weather(mosaicData)

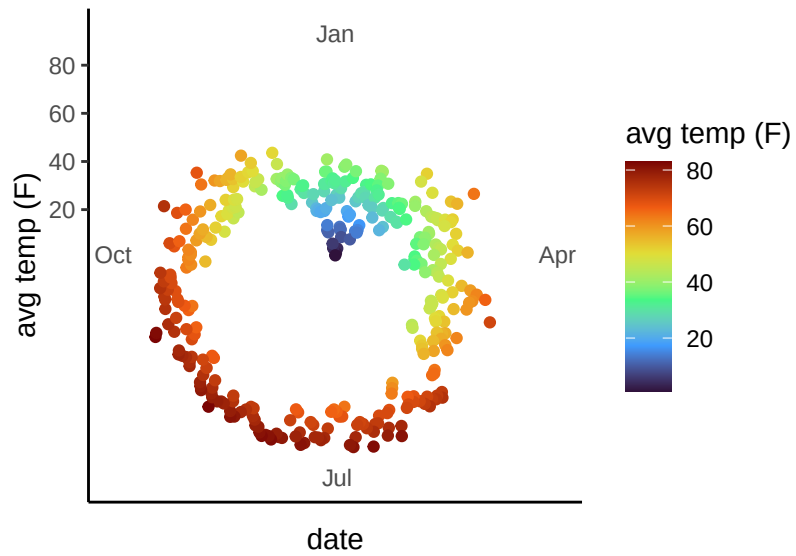
RADIAL TEMPERATURE PLOT SECTION

```
Chicago2016 <- subset(Weather, city == "Chicago" & year(date) == 2016)
```

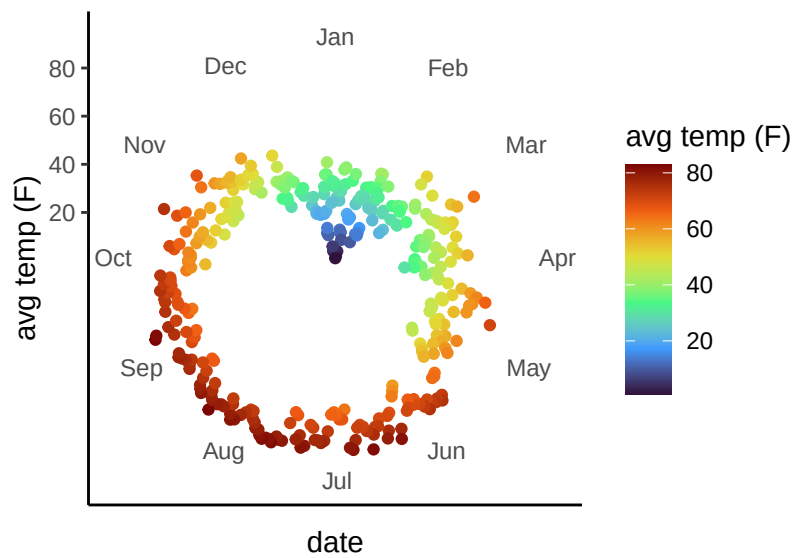
```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp)
```



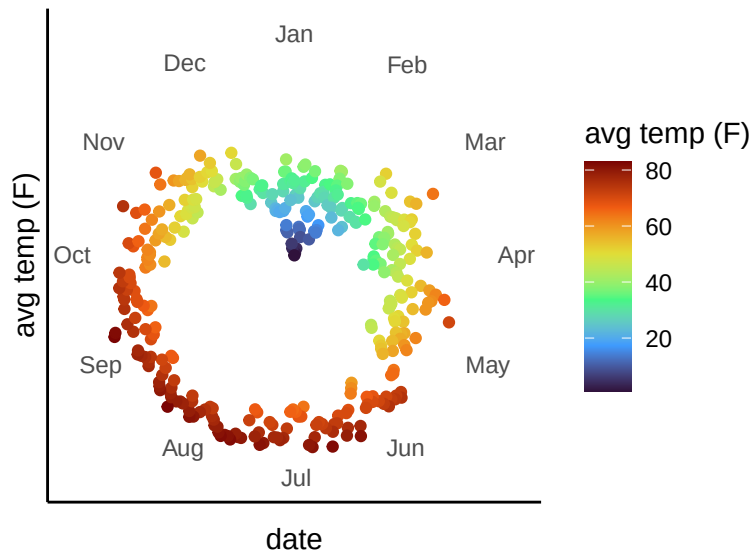
```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp) %>% gf_refine(scale_color_viridis_c(opt
```



```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp) %>%gf_refine(scale_color_viridis_c(opt
```



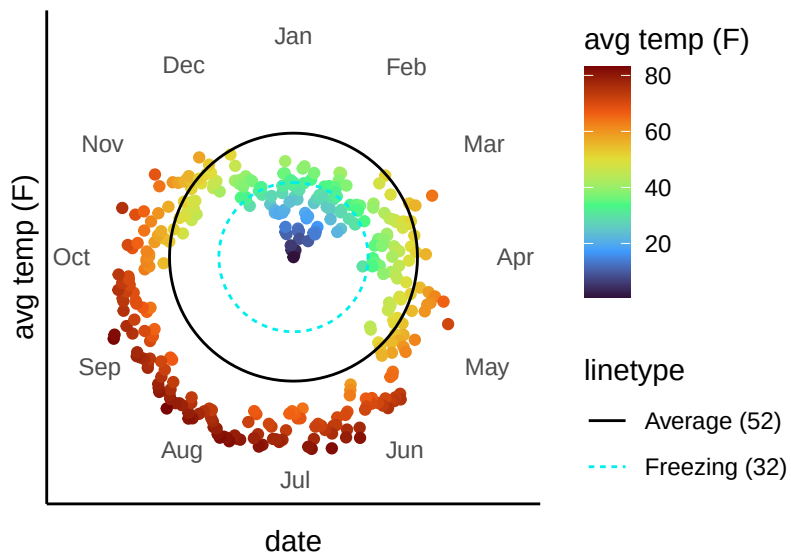
```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp) %>%gf_refine(scale_color_viridis_c(opt
```



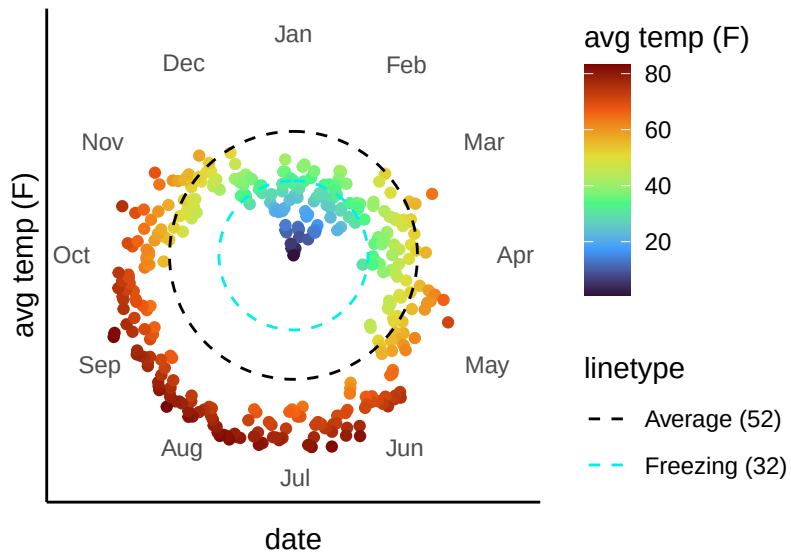
```
favstats(~avg_temp, data = Chicago2016)
```

```
## min Q1 median Q3 max mean sd n missing
## 1 37 54 71 83 52.54372 20.07594 366 0
```

```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp) %>%gf_refine(scale_color_viridis_c(opt
```



```
gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp) %>%gf_refine(scale_color_viridis_c(opt
```



```
#Final
#gf_point(avg_temp ~ date, data = Chicago2016, color = ~avg_temp)%>%gf_labs(title="January is Chicago's")
```

CONNECTED SCATTERPLOT SECTION

```
gf_path(high_temp ~ low_temp, data = subset(Weather, month == 5 & year == 2017 & city == "Chicago"), color = ~day)
```

Connected Scatterplot: High vs Low Temperatures
Chicago – May 2017

